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| REQ-MIL-STD-1553-CMD-01 | <b>CreateCommandWord</b><br><b>Prototype:</b> uint16_t CreateCommandWord( uint8_t rtAddress, uint8_t transmit, uint8_t subAddress, uint8_t wordCount)<br><b>Description:</b> The function shall create a 16-bit command word from the provided parameters.                                                                                                                                                                                                                                                 |
| REQ-MIL-STD-1553-CMD-02 | <b>IF</b> all of the following conditions are satisfied:<br>a. rtAddress, transmit, subAddress and wordCount parameters are provided<br>Following operations are performed:<br>a. place the wordCount field in Bits 0–4<br>b. place the subAddress field in Bits 5–9<br>c. place the T/R field in Bit 10<br>d. place the rtAddress field in Bits 11–15<br>e. return the resulting 16-bit command word                                                                                                      |
| REQ-MIL-STD-1553-CMD-03 | <b>DecodeCMD</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| REQ-MIL-STD-1553-CMD-04 | <b>Prototype:</b> DecodedCommand DecodeCMD(uint16_t cmd)<br><b>Description:</b> The function shall decode a 16-bit command word into a DecodedCommand structure.                                                                                                                                                                                                                                                                                                                                           |
| REQ-MIL-STD-1553-CMD-05 | <b>IF</b> all of the following conditions are satisfied:<br>a. cmd is a 16 bit Command Word.<br>Following operations are performed:<br>a. extract Bits 0–4 from cmd and assign the value to the wordCount field.<br>b. extract Bits 5–9 from cmd and assign the value to the subaddress field.<br>c. extract Bit 10 from cmd and assign the value to the transmit field.<br>d. extract Bits 11-15 from cmd and assign the value to the rtAddress field.<br>e. return the decoded DecodedCommand structure. |
| REQ-MIL-STD-1553-STA-01 | <b>CreateStatusWord</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| REQ-MIL-STD-1553-STA-02 | <b>Prototype:</b> uint16_t CreateStatusWord(uint8_t rtaddress, bool me)<br><b>Description:</b> The function shall create a 16-bit status word from the provided parameters.                                                                                                                                                                                                                                                                                                                                |
| REQ-MIL-STD-1553-STA-03 | <b>IF</b> all of the following conditions are satisfied:<br>a. rtAddress and me parameters are provided<br>Following operations are performed:<br>a. randomly generate the Busy field<br>b. set the Busy field to `1` with a probability of approximately 2% and to `0` otherwise<br>c. place the Busy field in Bit 9<br>d. place the Message Error field in Bit 10<br>e. place the RT Address field in Bits 11–15<br>f. return the resulting 16-bit status word                                           |
| REQ-MIL-STD-1553-STA-04 | <b>DecodeStatus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| REQ-MIL-STD-1553-STA-05 | <b>Prototype:</b> DecodedStatus DecodeStatus(uint16_t stat)<br><b>Description:</b> The function shall decode a 16-bit status word into a DecodedStatus structure.                                                                                                                                                                                                                                                            |
| REQ-MIL-STD-1553-STA-06 | <b>IF</b> all of the following conditions are satisfied:<br>a. stat is a 16 bit Status Word.<br>Following operations are performed:<br>a. extract Bit 9 from stat and assign the value to the busy field.<br>b. extract Bit 10 from stat and assign the value to the messageError field.<br>c. extract Bits 11-15 from stat and assign the value to the rtAddress field.<br>f. return the resulting DecodedStatus structure. |